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Knowledge graph enhanced recommendation via context-aware dynamic relation perception and dual-view hierarchical contrastive learning

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ABSTRACT

Knowledge graphs (KGs) provide a structured, multi-relational context to enhance recommender systems. However, existing KG-based methods face two critical challenges: (1) reliance on static entity embeddings, which fail to capture dynamic semantic shifts across different relational contexts, and (2) vulnerability to noisy or weakly relevant relations that propagate and impair model robustness. To address these issues, we propose CRHCL, a novel KG-enhanced recommendation framework that integrates context-aware relation modeling with a hierarchical contrastive learning strategy. First, we propose a context-aware relation modeling method that can dynamically disambiguate entity semantics across diverse relational contexts, enabling precise capture of context-dependent semantics. Second, we construct two complementary semantic views–one derived from user-item interaction behavior and the other from knowledge-guided propagation, and optimize them via a multi-level contrastive learning strategy. This strategy enforces both intra-layer consistency (within the same semantic layer) and inter-layer alignment (across propagation layers), enhancing robustness against noise. Extensive experiments on three public benchmark datasets validate CRHCL's superiority. It outperforms state-of-the-art baselines across metrics like AUC, F1-score, and Recall@K.

1. Introduction

Recommendation systems have become indispensable in delivering personalized content across diverse domains, from e-commerce platforms to streaming services and social networks (Bao et al., 2025; Chen et al., 2025; Deldjoo et al., 2024; Fan, 2024; Kim et al., 2025; Li et al., 2024a; Zangerle & Bauer, 2022). Traditional approaches, such as collaborative filtering (CF) (Hao et al., 2024) and matrix factorization (MF) (Zheng et al., 2022), focus on modeling user-item interaction histories to infer preferences. While effective in many scenarios, these methods suffer from critical limitations: they struggle with data sparsity and fail to capture high-order semantic relationships, especially when user behavior is sparse or preferences are complex.

To address these gaps, knowledge graphs (KGs) (Elahi et al., 2025; Liang et al., 2024; Meng et al., 2025; Pan et al., 2024; Zhang et al., 2025b) have emerged as a powerful tool to enrich recommendation systems. KGs provide structured, multi-relational auxiliary knowledge that connects entities (e.g., items, users, attributes) through explicit relations, enabling more nuanced modeling of user interests and item characteristics. Early efforts integrated pre-trained KG embeddings (e.g., TransE (Bordes et al., 2013), TransH (Wang et al., 2014)) with collaborative signals, but treated KG embedding and recommendation learning as separate stages, missing opportunities to model semantic dependencies in a unified framework (Sung et al., 2024; Zheng et al., 2024).

Recent advances have shifted toward leveraging multi-hop relational paths (Choi et al., 2024; Jin et al., 2021) or graph neural networks (GNNs) (Corso et al., 2024; Hu et al., 2025; Li et al., 2024b; Liu et al., 2024; Yin et al., 2024) to capture high-order correlations. GNN-based methods, in particular, have shown promise by jointly learning from user-item interaction graphs and KG subgraphs through multi-hop neighbor aggregation (Lai et al., 2024; Wang et al., 2023a). However, two fundamental challenges persist:

(1) Contextual Semantic Ambiguity: Most existing methods use static or globally shared entity representations, failing to account for

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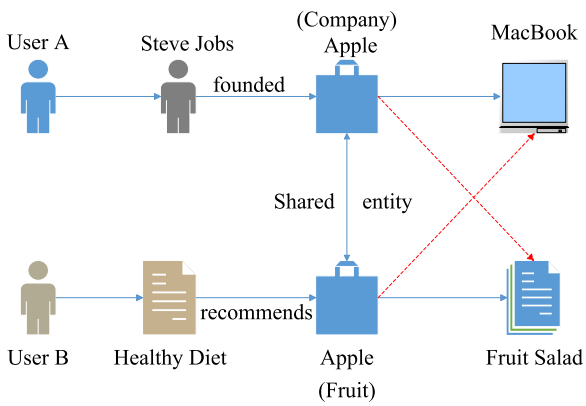


Fig. 1. Contextual semantics of polysemous entities: the meaning of "Apple" varies with user-specific relational paths.

semantic shifts under different relational contexts. For example, the entity "Apple" may refer to a technology company in one path (e.g., "Steve Jobs founded Apple") and a fruit in another (e.g., "Apple is an ingredient in fruit salad") (Fig. 1). Without context-aware modeling, such polysemy cannot be resolved, limiting the ability to infer fine-grained user intent.

(2) Noise Amplification in Multi-Hop Propagation: KGs—specially those automatically constructed—often contain noisy or weakly relevant triples. GNN-based multi-hop aggregation can amplify this noise, leading to semantic drift and degraded recommendation performance.

Meanwhile, contrastive learning (CL) (Huang et al., 2024; Sun et al., 2025; Xu et al., 2025) has emerged as a powerful self-supervised paradigm for enhancing representation robustness in recommendation systems. By aligning complementary views of data, CL mitigates issues like sparse supervision and noise interference. However, existing CL frameworks for knowledge-aware recommendation often rely on shallow contrastive objectives or neglect the hierarchical nature of semantic learning in multi-hop propagation, limiting their ability to handle relational ambiguity and noise.

Recent models such as KGCL (Yang et al., 2022) and CMCLRec (Xu et al., 2024) have incorporated CL into knowledge-enhanced recommendation. However, KGCL focuses primarily on graph-level contrastive alignment without contextual disambiguation, while CMCLRec addresses cold-start issues through cross-modal mapping, yet lacks graph-based semantic reasoning.

To overcome these limitations, we propose CRHCL–a novel KG-enhanced recommendation framework that synergizes Context-aware Relation modeling with Hierarchical Contrastive Learning. CRHCL dynamically disambiguates entity semantics based on relational paths while enforcing multi-level consistency between interaction and knowledge views, resulting in more precise and robust representations. Our contributions are summarized as follows:

- To overcome the limitations of static entity representations, we proposed a context-aware relation modeling method that adaptively refines entity representations based on their surrounding relational paths, effectively resolving semantic ambiguity for finer-grained reasoning.
- To avoid noise accumulation and performance degradation in multi-hop propagation, we designed a dual-view hierarchical contrastive learning strategy that aligns user-item interaction and KG propagation views at multiple semantic levels. This novel approach enhances robustness by enforcing intra-view consistency and cross-view alignment, thereby mitigating noise amplification.
- We integrate these two components into the CRHCL framework, demonstrating how dynamic semantic disambiguation and multi-level representation alignment can be jointly optimized for superior recommendation.
- We conduct extensive experiments on three public benchmarks (Book-Crossing, Last.FM, and MovieLens-1M). CRHCL consistently surpasses state-of-the-art baselines, achieving an average improvement of 0.78 % in AUC, 0.60 % in F1-score, and 1.17 % in Recall@K. These results, together with strong performance on the large-scale MovieLens-1M dataset, demonstrate the scalability and practical applicability of CRHCL in real-world scenarios.

The remainder of this paper is structured as follows. Section 2 reviews related work. Section 3 formalizes the problem. Section 4 describes the CRHCL framework in detail. Section 5 presents the experiments and analysis. Section 6 concludes the paper and discusses future directions.

2. Related work

2.1. Knowledge-aware recommendation

KGs have become a vital component in modern recommendation systems due to their ability to encode structured and semantically rich information about entities and their relations. By incorporating auxiliary knowledge, KGs enhance representation learning and improve the interpretability of recommendations. In recent years, this integration has led to the emergence of knowledge-enhanced recommendation models. These methods generally fall into three main categories based on how they leverage KG information: embedding-based approaches, path-based strategies, and GNN-based models.

2.1.1. Embedding-based methods

One representative line of early knowledge-aware recommendation methods leverages knowledge graph embedding (KGE) techniques, such as TransE (Bordes et al., 2013) and TransH (Wang et al., 2014), to learn low-dimensional representations of entities and relations in advance. These pre-trained embeddings are then incorporated into recommendation models, often in combination with collaborative filtering architectures. For example, CKE (Zhang et al., 2016) integrates structural knowledge, textual descriptions, and visual features within a unified Bayesian framework. RippleNet (Wang et al., 2018) further expands user preferences by propagating interactions along multi-hop relational paths in the KG.

Despite demonstrating the feasibility of incorporating structured knowledge into recommendations, these approaches exhibit several limitations. A primary issue is the separation between KG embedding learning and user-item interaction modeling. Since KGE models are generally trained for link prediction, their objectives are centered on preserving structural semantics within the graph. This is inherently different from the goal of recommendation, which emphasizes learning personalized preferences based on collaborative signals. As a result, the semantic representations derived from KGE often fail to align with user behavior patterns, limiting their effectiveness in capturing complex and dynamic recommendation scenarios.

2.1.2. Path-based methods

Path-based methods enhance recommendations by utilizing the semantic information embedded in multi-hop paths between users and items within the KG. These methods typically model the KG as a heterogeneous information network (HIN) and aim to capture higher-order connectivity by designing or mining meta-paths and meta-graphs. For instance, PER (Yu et al., 2014) employs handcrafted meta-path features to characterize user-item relationships. At the same time, KPRN (Wang et al., 2019d) automatically extracts multi-hop paths and models their sequential dependencies using recurrent neural networks (RNNs) for preference prediction.

By explicitly leveraging the topological structure of KGs, path-based approaches offer improved interpretability and provide semantically meaningful connection patterns. However, their performance is highly

sensitive to the quality of the selected meta-paths or meta-graphs. Manually designing effective paths often requires significant domain expertise and considerable engineering effort, especially in large-scale KGs with complex relational schemas. Furthermore, handcrafted paths may fail to capture implicit or diverse interaction patterns, limiting their generalization across different datasets or domains. These limitations constrain the scalability and adaptability of path-based methods in practical recommendation scenarios.

2.1.3. GNN-based methods

In recent years, GNNs have emerged as the predominant approach for KG representation, owing to their ability to model graph-structured data and aggregate multi-hop neighborhood information. Typically, these methods integrate user-item interaction graphs with KGs into a unified heterogeneous structure—such as a user-item-entity graph—and apply GNN variants like Graph Convolutional Networks (GCNs) or Graph Attention Networks (GATs) (Ying et al., 2018) to learn user and item representations through iterative message passing. For example, KGCN (Wang et al., 2019b) and KGNN-LS (Wang et al., 2019a) employ GCNs on item-side KGs to enrich item embeddings. KGAT (Wang et al., 2019c) utilizes GATs over unified graphs to capture high-order semantic signals. More advanced models, such as KGIN (Wang et al., 2021a), explicitly model user intents through a user-intent-item-entity structure, while CG-KGR (Chen et al., 2022) leverages collaborative signals to guide KG aggregation. COAT (Dai et al., 2022) simultaneously models both user-item and KGs by combining GNNs with personalized attention to capture collaborative and semantic patterns.

Despite their effectiveness, these GNN-based methods still struggle with contextual semantic ambiguity and noise accumulation, which motivate the design of our CRHCL framework.

2.2. Contrastive learning

In recent years, CL (Huang et al., 2024; Sun et al., 2025; Sung et al., 2024; Zheng et al., 2024) has emerged as a powerful self-supervised learning paradigm in knowledge-aware recommendation (Zhang et al., 2025a). By constructing positive and negative pairs without relying on labeled data, CL can effectively enhance the discriminative ability and robustness of representations. It offers promising solutions to several long-standing challenges in GNN-based KGR models, including sparse supervision, imbalanced information propagation, and noise interference from irrelevant knowledge.

Early efforts in CL for recommendation focus on user-item interaction graphs. For instance, SGL (Wu et al., 2021) adopts structural augmentations such as edge and node dropout to generate multiple graph views and applies CL at the graph level. NCL (Lin et al., 2022) extends this idea by incorporating node-level contrast with latent relevance estimation, enabling more fine-grained representation learning. In knowledge-aware scenarios, CL further facilitates semantic modeling by aligning different relational structures. MCCLK (Zou et al., 2022a) introduces a multi-level cross-view contrastive framework that incorporates global and local views to capture collaborative and semantic signals simultaneously. KGCL (Yang et al., 2022) enforces consistency across subgraph structures through graph-level contrastive objectives, improving entity-level discrimination under limited supervision.

While KG-enhanced CL has demonstrated promising results, existing approaches differ substantially from our design. KGCL primarily imposes contrastive regularization by constructing alternative structural views, such as node-level or neighbor-level perturbations, to suppress propagation noise. However, it does not explicitly encode relational path semantics or enforce consistency across multi-hop semantic hierarchies. CMCLRec, on the other hand, focuses on the cold-start scenario by performing cross-modal alignment between user profile modalities and interaction behaviors, and does not address relational ambiguity or cross-layer semantic alignment in KGs.

Table 1
Explanation of formula symbols.

Symbol	Explanation
U	Set of users
I	Set of items
Y	User-item interaction matrix
$\mathcal{G}$	KG with triples
$\mathfrak{I}$	Set of entities
$\mathfrak{R}$	Set of relations
S	Set of ripple
T	Set of item-entity
e_h	Head entity
r	Relations
e_t	Tail entity
$\hat{y}_{u,i}$	Prediction function
$\mathfrak{I}_u^0$	User's initial entity set
$U_{potential}$	User's potential entity sets
$I_{potential}$	Item's potential entity sets
$\mathfrak{I}_i^0$	Item's initial entities set
$\mathfrak{I}_{o,L}^l$	Entity set of target o at layer l
$S_{o,L}^l$	Ripple set of target o at layer l
$\mathcal{N}_{e_h}^*$	Sum of head entities
H_l	Connection of relationship paths
$\pi(\cdot)$	Function to calculate tail entity attention weights
α	Tail entity embedding attention weight
$\tilde{e}$	Augmented embedding
agg_{concat}^o	Connection-based aggregator
$\mathcal{L}$	Loss function

In contrast to prior KG-based contrastive models, our framework integrates relation-aware semantic modeling with hierarchical contrastive learning in a unified manner. Rather than treating KG neighbors uniformly or constructing only structural perturbations, our model adaptively assigns semantic importance based on relation-path context and enforces consistency both within and across propagation layers. This enables fine-grained semantic disambiguation and stable multi-hop reasoning, providing complementary benefits beyond conventional KGCL-style propagation regularization and CMCLRec-style modality alignment.

3. Problem formulation

In this section, we first provide a brief introduction to the relevant concepts, including the user-item interaction graph and the definition of the KG. The mathematical formula symbols used in this paper are defined in Table 1. We then describe the specific details of the recommendation task.

User-Item Interaction Graph: Let $U = [u_0, u_1, u_2, \dots, u_m]$ denote the set of users and $I = [i_0, i_1, i_2, \dots, i_n]$ denote the set of items, where m and n represent the total number of users and items, respectively. Based on implicit user feedback (e.g., clicks, views, or purchase records), a user-item interaction matrix $Y \in R^{m\times n}$ can be constructed, where:

$$Y_{u,i} = \begin{cases} 1, & \text{if user-item interaction } (u,i), \\ 0, & \text{otherwise}, \end{cases} \tag{1}$$

where $Y_{u,i} = 1$ indicates that the user has expressed interest in the item, as reflected by behaviors such as viewing, reading, or purchasing, otherwise, $Y_{u,i} = 0$ indicates no observed interaction.

Knowledge Graph: A KG is a graph-based data structure designed to represent real-world entities and the semantic relationships between them. Thus, a KG can be organized as $\mathcal{G} = \{(e_h, r, e_t) | e_h, e_t \in \mathfrak{I}, r \in \mathfrak{R}\}$, where e_h, r, and e_t serve as representations of the head, relation, and tail of a knowledge triple, correspondingly. $\mathfrak{I}$ and $\mathfrak{R}$ represent the sets of entities and relations contained within $\mathcal{G}$. Across many recommendation contexts, an item $i \in I$ corresponds to one specific entity $e \in \mathfrak{I}$. To capture this connection, we establish a set of item-entity alignments $T = \{(i, e) \mid i \in I, e \in \mathfrak{I}\}$, where the tuple (i, e) denotes that item i can be matched to entity e in the KG.

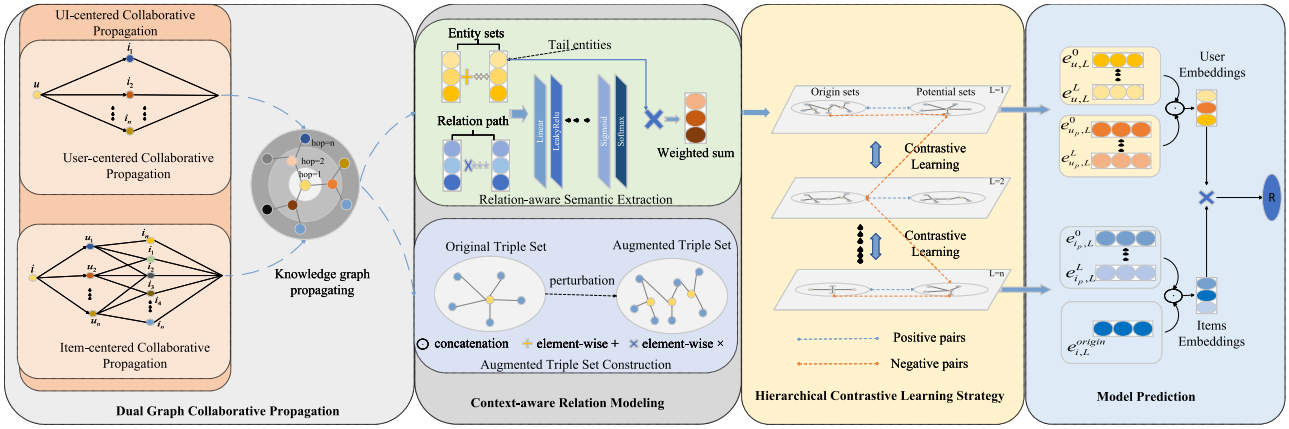


Fig. 2. The overall architecture of the proposed CRHCL framework. The framework comprises dual graph collaborative propagation over the interaction and KGs, context-aware relation modeling, hierarchical contrastive learning strategy, and model prediction, which is generated by aggregating representations across both semantic views.

Recommendation Task: Given a user-item interaction matrix Y and a KG $\mathcal{G}$, the goal of the knowledge-aware recommendation task is to build a model that learns a function $\hat{y}_{u,i} = \mathcal{F}(u, i|\mathcal{G}, \Theta)$, where Θ denotes the model parameters of function $\mathcal{F}$. This function aims to predict the likelihood that a user will interact with a given item. Specifically, the model analyzes the user's historical interaction data along with rich semantic information from the KG, enabling it to more accurately capture user preferences and interests and ultimately generate personalized recommendations.

4. Methodology

The proposed CRHCL framework addresses the limitations of existing KG-enhanced recommendation methods by integrating context-aware dynamic relation perception and dual-view hierarchical contrastive learning. As shown in Fig. 2, the framework consists of four core components, designed to sequentially resolve semantic ambiguity, suppress noise propagation, and enforce cross-view consistency: **(1) Dual Graph Collaborative Propagation Layer:** This module combines the user-item interaction graph and the KG to propagate high-order semantic signals from both sources. By leveraging the complementary structures of these graphs, it effectively captures the intricate relationships between user preferences and item attributes from dual perspectives, enriching the representation of user-item interactions. **(2) Context-aware Relation Modeling Layer:** To enhance semantic accuracy, this layer models the contextual semantics of entities by considering their interactions within different relational paths. It dynamically adjusts entity representations based on their specific contexts in the KG, ensuring that polysemous entities are represented according to their distinct meanings in varying relational contexts. This enables precise disambiguation of entity semantics across diverse paths, thereby improving the accuracy of semantic representation. **(3) Hierarchical Contrastive Learning Strategy Layer:** Building upon the collaborative propagation mechanism, this component introduces a hierarchical contrastive strategy that promotes semantic consistency across multiple layers. By aligning representations within each layer and across different layers, this framework ensures robust feature learning, reduces the impact of irrelevant or redundant information, and stabilizes the model's predictions in complex KG environments. **(4) Model Prediction Layer:** The final layer aggregates the multi-layer representations of users and items into unified embeddings. These embeddings are then used to compute interaction probabilities, generating the final recommendation results based on the aggregated semantic signals from both the user-item interaction data and the KG.

4.1. Dual graph collaborative propagation layer

The Dual Graph Collaborative Propagation Layer serves as the foundation for extracting multi-hop semantic signals from both the user-item (UI) interaction graph and the KG. It comprises two complementary sub-layers: one dedicated to propagating collaborative preferences through user-item interactions, and the other focused on structural semantics embedded in the KG.

This layer enables the model to integrate behavioral and semantic signals in a unified representation space. By recursively aggregating information from neighboring nodes in both graphs, the model captures high-order dependencies that reflect users' implicit interests and the semantic relevance of items. The collaborative sub-layer enhances the expressiveness of user-item relationships, while the knowledge-based sub-layer expands entity semantics through multi-hop relational paths. Together, they provide a comprehensive foundation for downstream semantic alignment and consistency in representation.

The following subsections detail the mechanisms of these two propagation sub-layers and how they interact to produce context-enhanced user and item embeddings.

4.1.1. UI-centered collaborative propagation sub-layer

User-centered collaborative propagation: In traditional recommendation systems, user representations are typically modeled as static latent vectors, which limits the model's ability to capture the complex and context-specific nature of user preferences. To overcome this limitation, we propose a user-centered collaborative propagation mechanism that incorporates both user-item interactions and KG semantics into a unified representation framework.

Specifically, we first establish direct propagation paths between each user and their historically interacted items. These interactions provide a behavioral foundation for modeling user interests. To enrich the semantic granularity of this process, we further integrate knowledge-level information by extracting associated entities and relations from the KG for each interacted item. This enables the model to gather multi-hop contextual signals and generate more informative user embeddings.

We begin by constructing the user's initial entity-level set, denoted as $\mathfrak{I}_u^0$, based on the user's positively interacted items and their corresponding KG neighbors. Formally, this set is defined as:

$$\mathfrak{I}_u^0 = \left\{ e|(i,e) \in T,\ y_{u,i} = 1 \right\}, \tag{2}$$

where T denotes the set of item-entity mapping and $y_{u,i} = 1$ indicates an observed interaction between user u and item i.

To further enhance the user representation, we introduce a potential user set $U_{potential}$ that captures users with similar interaction patterns.

These users are identified based on shared item preferences, thereby enabling the aggregation of semantically relevant entities beyond the user's immediate history. The potential user set is formally defined as:

$$U_{potential} = \left\{ u_p | \exists i,\ y_{u,i} = 1 \wedge y_{u_p,i} = 1 \right\}, \tag{3}$$

where u_p denotes potential users. By aggregating both the initial and potential user representations across multiple hops and semantic layers, the model can effectively learn fine-grained, context-aware user embeddings that reflect not only explicit preferences but also latent behavioral semantics.

Item-centered collaborative propagation: Users who interact with the same item often exhibit correlated or overlapping preferences. We can leverage these collective user behaviors to obtain a valuable signal for refining item-level representations. To capture this effect, we propose an item-centered collaborative propagation mechanism that aggregates user engagement patterns to uncover implicit relationships among items.

Specifically, for each target item i, we identify a group of users U_i who have interacted with it. By analyzing the interaction histories of these users, we discover other items that co-occur with i through shared user behaviors. These co-engaged items reflect semantically similar characteristics from the perspective of collaborative preferences. Formally, the potential item set associated with item i is defined as:

$$I_{\text{potential}} = \left\{ i_u | \exists u \in U,\ y_{u,i} = 1 \wedge y_{u,i_u} = 1 \right\}, \tag{4}$$

where $y_{u,i} = 1$ indicates an observed interaction between user u and item i, and U denotes the complete set of users.

To further enrich the semantic context of the item, we extract knowledge-level information from the KG for all items in $I_{\text{potential}}$. In particular, we collect entities directly connected to these items to form the initial entity set of each item. This set is defined as:

$$\mathfrak{I}_i^0 = \left\{ e | (i_u, e) \in T,\ i_u \in I \right\}, \tag{5}$$

where T denotes the set of entity–relation triples in the KG.

The proposed user-centered and item-centered propagation sublayers serve as complementary components within the CRHCL framework. While the former focuses on extracting personalized semantic signals by aggregating knowledge-enriched interactions around a user's behavior history, the latter uncovers cross-item semantic associations through shared user engagement. By jointly modeling collaborative signals from both user and item perspectives, the framework constructs semantically enhanced and behavior-aware representations.

This dual propagation strategy not only improves the expressiveness of learned embeddings but also contributes to better recommendation accuracy, interpretability, and generalization in the presence of sparse data or relational ambiguity.

4.1.2. Knowledge graph propagation sub-layer

KG propagation: In KGs, entities connected by relational edges often share strong semantic correlations. To leverage this property, we incorporate a ripple-based knowledge propagation mechanism inspired by Wang et al. (2020b). This module enables the model to capture high-order semantic dependencies by iteratively expanding from the original entities along relational paths.

Starting from the initial entity sets generated by collaborative propagation, the model recursively explores multi-hop neighborhoods in the KG. At each layer, it aggregates relational triples connected to the current set of entities, thereby forming a ripple-like expansion structure. This process allows the model to enrich user and item representations with layered semantic context, progressively building a more expressive relational profile.

Formally, let $\mathcal{G}$ denote the KG set, and let $o \in \{u, i\}$ be a placeholder symbol representing either a user (u) or an item (i). We define the lth layer entity set for target o as:

$$\mathfrak{I}_{o,L}^l = \left\{ e_t | (e_h, r, e_t) \in \mathcal{G} \text{ and } e_h \in \mathfrak{I}_{o,L}^{l-1} \right\}, \quad l = 1, 2, \ldots, L, \tag{6}$$

where L is a hyperparameter indicating the maximum number of propagation layers (i.e., hops), and $\mathfrak{I}_{o,L}^{l-1}$ represents the entity set from the previous layer. Each layer expands the entity set by collecting tail entities reachable via any relation from entities in the preceding layer.

Correspondingly, the set of relational triples, or ripple set, extracted at layer l is defined as:

$$S_{o,L}^l = \left\{ (e_h, r, e_t) | (e_h, r, e_t) \in \mathcal{G} \text{ and } e_h \in \mathfrak{I}_{o,L}^{l-1} \right\}, \quad l = 1, 2, \ldots, L \tag{7}$$

Through this ripple-based propagation, the model is able to traverse semantically meaningful paths originating from the user's or item's initial neighborhood. This process introduces informative and structurally coherent context, which complements the behavior-driven and collaborative signals. Ultimately, the resulting hierarchical entity and relation sets serve as a rich foundation for downstream semantic alignment and contrastive representation learning.

4.2. Context-aware relation modeling layer

Entities in a KG often participate in multiple relational paths, leading to semantic ambiguity and context-dependent meaning shifts. For instance, the same entity may represent different concepts when appearing in different neighborhoods or under different relation types. Capturing such variation is critical for learning expressive and personalized representations in recommendation tasks.

To address this, we design a context-aware relation modeling layer that explicitly models the semantic dependencies between entities and their relational context, while also introducing controlled perturbations to simulate semantic variation. This module consists of two complementary components:

- **Relation-aware semantic extraction**, which dynamically adjusts entity embeddings based on their relational paths using a relation-context encoding function. This allows the model to disambiguate polysemous entities and capture path-specific semantics.
- **Augmented triple set construction**, which generates alternative semantic views of entities by applying perturbations to the original triple structures. These views enhance the model's robustness and support CL objectives.

Together, these components enable the model to build semantically aligned representations that are both context-sensitive and noise-resilient. In the following, we detail each component.

4.2.1. Relation-aware semantic extraction

In KGs, each item is connected to various entities through direct or indirect relational paths. These associated entities act as critical semantic descriptors, offering valuable signals to uncover user preferences. For instance, the appeal of music albums like Thriller or The Dark Side of the Moon may stem from associations with entities such as Michael Jackson or the rock genre. To effectively model such structured knowledge, we propagate relevant entity information along multi-hop relational paths and integrate it into the representation learning process.

To address semantic polysemy, we introduce a relation-aware semantic extraction mechanism that adaptively disambiguates tail entities based on their relational contexts. Following Wang et al. (2020b), we define a contextual path consisting of previously visited head entities and relations. Let $(e_h, r, e_t) \in S_{o,L}^l$ denote a triple in the lth layer, where e_h represents the head entity in the KG triple, r denotes the relation connecting the head entity e_h and the tail entity e_t, and e_t stands for the tail entity in the KG triple. We represent the relational history and corresponding entity path as:

$$\mathcal{N}_{e_h} = e_h^1 \oplus e_h^2 \oplus \cdots \oplus e_h^{l-1}, \quad H_l = r_1 \circ r_2 \circ \ldots \circ r_{l-1}, \quad l = 1, 2, \ldots, L, \tag{8}$$

where $\oplus$ and $\circ$ denote element-wise sum and relation path concatenation, respectively.

$$\alpha = \pi(e_h, N_{e_h}, r_l, H_l) \tag{9}$$

We compute the attention weights α for tail entity embeddings using a neural relation-context encoding function $\pi(\cdot)$, which captures the importance of each tail entity under specific relational contexts. To implement this function, we design a multi-layer feedforward network that dynamically computes context-aware attention weights over relational paths. The architecture of this network is motivated by three key considerations:

- Non-linear Relationship Modeling: The interactions between entities and relations along complex paths are inherently non-linear. Simple linear models (e.g., dot products) cannot adequately capture this complexity. Thus, we employ a multi-layer perceptron (MLP) that leverages stacked non-linear transformations to learn intricate interaction functions among entities, relations, and their historical paths.
- Feature Fusion and Dimensionality Reduction: The network input is a concatenated vector of the head entity, relational history, and current relation—high-dimensional and heterogeneous. The first layer (W_0, b_0) acts as a feature fusion and dimensionality reduction layer, projecting inputs into an expressive low-dimensional space using LeakyReLU activation to preserve gradient information.
- Stable and Robust Training: We integrate Dropout and LayerNorm to address KG challenges like data sparsity and distribution instability. Dropout prevents overfitting to specific entities or relations, while LayerNorm stabilizes intermediate features and accelerates convergence—crucial for mitigating gradient issues in multi-hop propagation.

This function is implemented as a multi-layer feedforward network (Vaswani et al., 2017), which use Drop to indicate dropout (Liu et al., 2023b), LeakyReLU activation, and Layer Normalization (De Min et al., 2023):

$$z = \mathrm{Norm}(W_1 \cdot \mathrm{Drop}(\mathrm{LeakyReLU}(W_0 \cdot ((e_h \oplus \mathcal{N}_{e_h}) \parallel (r_l \otimes H_l)) + b_0)) + b_1) + b_2 \tag{10}$$

$$\pi(e_h, N_{e_h}, r_l, H_l) = \sigma(W_2 \cdot \mathrm{LeakyReLU}(z)), \tag{11}$$

where $\sigma(\cdot)$ denotes the sigmoid function (Elfwing et al., 2018), and $\parallel$ denotes vector concatenation, and $\otimes$ represent element-wise product. W (weight matrices) and b (biases) are trainable parameters, here, distinct subscripts correspond to the parameters of different layers, respectively. To ensure proper relevance weighting across triples, we normalize the contextual importance score using the softmax function (Liu et al., 2023a):

$$\pi(e_h, N_{e_h}, r_l, H_l) = \frac{\exp(\pi(e_h, N_{e_h}, r_l, H_l))}{\sum_{(e'_h, r', e'_t) \in S^l_{o,L}} \exp(\pi(e_h, N_{e_h}, r_l, H_l))} \tag{12}$$

The final context-aware representation for the lth layer is obtained by aggregating tail entities weighted by their relation-context relevance scores:

$$e^l_{o,L} = \sum_{e^o_t \in \mathfrak{I}^0_o} \pi(e_h, r_l, H_l) \cdot e^o_t, \quad l = 1, 2, \ldots, L \tag{13}$$

With this design, the feedforward network acts as a relational context-aware module. It transforms input features, exploits non-linear modeling capacity, and uses regularization to highlight discriminative semantic signals along multi-hop paths. In this way, it directly helps resolve contextual semantic ambiguity in KG-enhanced recommendation.

We also compute a representation for the initial entity set $\mathfrak{I}^0_o$, which serves as the root-level semantics for the user or item:

$$e_{o,0} = \frac{\sum_{e \in \mathfrak{I}^0_o} e}{|\mathfrak{I}^0_o|} \tag{14}$$

In contrast to users, items feature interrelated entities as their primitive form of representation, while users do not. Specifically, an entity primitively linked to an item is the entity in semantic space A that maps to the item itself, and it is defined as:

$$e^{\text{origin}}_{i,L} = \frac{\sum_{e \in \{e \mid (e,i) \in T\}} e}{|\{e \mid (e,i) \in T\}|} \tag{15}$$

Finally, we collect the complete set of triple-based semantic representations used in subsequent modeling:

$$\begin{aligned} \mathcal{J}_{u_{\text{init}}} &= \{e^0_{u,L}, e^1_{u,L}, \ldots, e^L_{u,L}\} \\ \mathcal{J}_{u_{\text{potential}}} &= \{e^0_{u_p,L}, e^1_{u_p,L}, \ldots, e^L_{u_p,L}\} \\ \mathcal{J}_{i_{\text{origin}}} &= \{e^{\text{origin}}_{i,L}\} \\ \mathcal{J}_{i_{\text{potential}}} &= \{e^0_{i_p,L}, e^1_{i_p,L}, \ldots, e^L_{i_p,L}\} \end{aligned} \tag{16}$$

These sets form the basis for contrastive representation learning, capturing multi-hop contextual semantics and supporting robust user-item matching.

4.2.2. Augmented triple set construction

To improve the semantic expressiveness and robustness of user and item representations, CRHCL incorporates an augmented triple construction mechanism to support CL. This module builds structurally diverse but semantically related views of each entity, enabling the model to better distinguish informative patterns and suppress noise. The design is inspired by recent advances in CL for graph representation (Xu et al., 2025; Zheng et al., 2024).

Specifically, for each user u and item i, we construct two categories of triple-based semantic views from the KG: (1) original propagation triples and (2) perturbed augmented triples. Together, these views form the basis for intra- and inter-layer contrastive optimization.

Original propagation triples.. We begin by sampling multi-hop triples from the ripple sets constructed in Section 4.1. Formally, the original triple set for target $o \in \{u, i\}$ is defined as:

$$e^l_{o,L} = \left\{ (e^{l,k}_h, r^{l,k}, e^{l,k}_t) \right\}^{L,K}_{l=1,k=1}, \tag{17}$$

where L denotes the number of ripple layers, and K is the number of triples sampled per layer l. These triples are directly extracted from the original KG $\mathcal{G}$ and encode the high-order semantics around the target node.

Augmented views for contrastive learning.. To construct a perturbed view, we introduce controlled semantic variations into the original triples. Specifically, for each original triple $(e^{l,k}_h, r^{l,k}, e^{l,k}_t)$, we generate its augmented counterpart $(\tilde{e}^{l,k}_h, \tilde{r}^{l,k}, \tilde{e}^{l,k}_t)$ by injecting noise or applying structure-preserving transformations:

$$e^{l'}_{o,L} = \left\{ (\tilde{e}^{l,k}_h, \tilde{r}^{l,k}, \tilde{e}^{l,k}_t) \right\}^{L,K}_{l=1,k=1} \tag{18}$$

Each embedding in the augmented view is obtained by applying a Gaussian perturbation to its corresponding original vector:

$$\tilde{\mathbf{e}} = \mathbf{e} + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2 \mathbf{I}), \tag{19}$$

where $\mathbf{e} \in \{e_h, r, e_t\}$ denotes the original embedding, and ϵ is sampled from a multivariate Gaussian distribution with zero mean and variance σ^2.

This augmentation strategy generates two structurally correlated yet semantically distinct views of each entity's local neighborhood. It encourages the model to learn discriminative features that remain stable across input perturbations. When combined with the hierarchical contrastive objectives described in Section 4.3, these augmented triples significantly enhance representation robustness and improve recommendation performance, particularly under noisy or sparse conditions.

4.3. Hierarchical contrastive learning strategy layer

Traditional embedding-based methods struggle to capture high-order semantic dependencies and often propagate noise across multi-hop reasoning paths in large-scale KGs. To address this challenge, we design a Hierarchical Contrastive Learning (HCL) strategy that enforces representation consistency from two complementary perspectives: (1) within a propagation layer to compact local semantics, and (2) across layers to maintain coherent multi-hop abstractions. Here, $e^0_{o,l}$ is the anchor representation of user/item o at layer l, $e^k_{o,l}$ represents positive samples from the same layer, and $e^{k'}_{o,l}$ $(k' \neq l)$ are negative samples from other layers. The temperature τ controls the sharpness of contrastive separation.

Positive/negative sampling strategy: CRHCL constructs contrastive supervision based on semantic proximity in the propagation hierarchy. Entities within the same hop-layer naturally share comparable semantic granularity (e.g., co-genre books or shared-topic relations), thus they form the positive set. Entities from other hop-layers represent weaker or more abstract semantics and may introduce noise, so they form the negative set. This design encourages local contextual consistency while preventing cross-layer semantic drift. For instance, if a user interacts with *The Hobbit*, then other fantasy novels at the same layer act as positives, whereas high-hop entities such as author biographical works or publisher nodes act as negatives.

4.3.1. Intra-layer contrastive learning

We treat the layer-l embedding of a user or item as an anchor and pull it toward neighbors within the same layer (local positives), while pushing it away from representations from other layers (cross-layer negatives). This yields compact, context-specific semantic clusters and helps resolve entity polysemy. Mathematically, the intra-layer contrastive loss for a user u at layer l is:

$$\mathcal{L}^U_{\text{Intra}} = \sum_{u\in U} -\log \frac{\sum_{k\in l} \exp\left(\frac{e^0_{u,l}\cdot e^k_{u,l}}{\tau}\right)}{\sum_{k\in l} \exp\left(\frac{e^0_{u,l}\cdot e^k_{u,l}}{\tau}\right) + \sum_{k'>l} \exp\left(\frac{e^0_{u,l}\cdot e^{k'}_{u,l}}{\tau}\right)} + \sum_{u\in U} -\log \frac{\sum_{k\in l} \exp\left(\frac{e^0_{u_p,l}\cdot e^k_{u_p,l}}{\tau}\right)}{\sum_{k\in l} \exp\left(\frac{e^0_{u_p,l}\cdot e^k_{u_p,l}}{\tau}\right) + \sum_{k'>l} \exp\left(\frac{e^0_{u_p,l}\cdot e^{k'}_{u_p,l}}{\tau}\right)} \tag{20}$$

This layer-wise constraint emphasizes fine-grained contextual meaning and improves robustness under sparse or noisy local interactions. Following the same principle, we compute $\mathcal{L}^I_{\text{Intra}}$ for items.

4.3.2. Inter-layer contrastive learning

Local signals alone may be insufficient for capturing abstract semantics. Thus, for each representation at layer k, we align it with its multi-hop aggregated counterpart at the same semantic level while treating mismatched depths as negatives. This encourages global semantic coherence and mitigates noise accumulated during deep propagation. The inter-layer contrastive loss for users is:

$$\mathcal{L}^U_{\text{Inter}} = \sum_{u\in U}\sum_{k\in l} -\log \frac{\exp\left(\frac{e^k_{u,l}\cdot e^k_{u_p,l}}{\tau}\right)}{\exp\left(\frac{e^k_{u,l}\cdot e^k_{u_p,l}}{\tau}\right) + \sum_{k'\neq k} \exp\left(\frac{e^k_{u,l}\cdot e^{k'}_{u_p,l}}{\tau}\right)} \tag{21}$$

This cross-layer alignment injects high-order knowledge while filtering off-context paths, enabling stable hierarchical reasoning. Analogously, we define $\mathcal{L}^I_{\text{Inter}}$ for items.

4.3.3. Overall contrastive loss

We sum user/item losses at both levels to obtain the final objectives:

$$\mathcal{L}_{\text{Intra}} = \mathcal{L}^U_{\text{Intra}} + \mathcal{L}^I_{\text{Intra}} \tag{22}$$

$$\mathcal{L}_{\text{Inter}} = \mathcal{L}^U_{\text{Inter}} + \mathcal{L}^I_{\text{Inter}} \tag{23}$$

Through simultaneous optimization of intra- and inter-layer objectives, CRHCL learns context-aware and noise-resilient representations that preserve local semantic fidelity and global relational structure.

4.4. Model prediction layer

During the propagation process within the relation-context encoding network, the representation at each layer encodes different semantic aspects of user and item behaviors. Lower layers typically capture local preference patterns, while higher layers focus on structural relationships and long-range dependencies. To effectively integrate these multi-level signals, we adopt a concatenation-based aggregation scheme inspired by GraphSAGE (Hamilton et al., 2017), which has shown strong performance in multi-hop representation learning.

Specifically, for a given user or item $o \in \{U, I\}$, we concatenate the semantic representations across all L layers and apply a linear transformation followed by a non-linear activation:

$$\text{agg}^o_{\text{concat}} = \sigma\left(\mathbf{W}_a \cdot \left[e^1_{o,L} \parallel e^2_{o,L} \parallel \cdots \parallel e^L_{o,L}\right] + \mathbf{b}_a\right), \tag{24}$$

where $\parallel$ denotes vector concatenation, $\sigma(\cdot)$ is the sigmoid activation function, and $\mathbf{W}_a$, $\mathbf{b}_a$ are learnable weight parameters. This process produces a unified representation $\text{agg}^o_{\text{concat}}$ that captures semantic signals from different hierarchical levels.

Once the user and item embeddings are obtained from both the original and potential semantic views, we compute the final representation for each by concatenating their respective components:

$$e_u = [e_{u_i} \parallel e_{u_p}], \quad e_i = [e_{i_{\text{origin}}} \parallel e_{i_p}], \tag{25}$$

where e_{u_i} and $e_{i_{\text{origin}}}$ denote the aggregated representations from the original user-item interactions and entity-level item propagation, while e_{u_p} and e_{i_p} capture complementary information from potential user and item views introduced during collaborative propagation.

Based on these final embeddings, the model estimates a matching score $\hat{y}_{u,i}$ between user u and item i using an inner product:

$$\hat{y}_{u,i} = e_u^\top e_i \tag{26}$$

This score reflects the user's preference strength for the item and is used during training and inference to rank candidate items. The design enables the model to preserve rich semantic dependencies while aligning both local and global representations into a unified interaction space.

4.5. Training process and optimization

To jointly learn informative user and item representations, the proposed model utilizes a multi-task learning framework and leverages self-supervised signals from the KG. It integrates both supervised recommendation signals and auxiliary contrastive objectives. Specifically, two types of CL losses—namely, intra-layer and inter-layer contrastive losses—are introduced to promote semantic consistency within and across semantic levels.

For the main recommendation objective, we employ the pairwise Bayesian Personalized Ranking (BPR) loss (Rendle et al., 2012), which encourages the model to assign higher scores to items that users have interacted with than to unobserved ones. Formally, the BPR loss is defined as:

$$\mathcal{L}_{BPR} = \sum_{(u,i)\in\rho^+} \sum_{(u,i)\in\rho^-} -\ln\sigma\left(\hat{y}_{u,i} - \hat{y}_{u,i}\right), \tag{27}$$

where ρ^+ and ρ^- denote the sets of observed and unobserved user-item interactions, respectively, and $\sigma(\cdot)$ is the sigmoid function.

To prevent overfitting and suppress noise propagation from the KG, we add a regularization term over all entity and relation embeddings:

$$\mathcal{L}_{regular} = \sum_{e_h\in\mathfrak{I}} \|e_h\|_2^2 + \sum_{e_t\in\mathfrak{I}} \|e_t\|_2^2 + \sum_{r\in\mathfrak{R}} \|r\|_2^2, \tag{28}$$

where $\mathfrak{I}$ and $\mathfrak{R}$ denote the sets of entities and relations in the KG, respectively.

The overall CL loss is the sum of the intra-layer and inter-layer objectives:

$$\mathcal{L}_{KG} = \mathcal{L}_{Intra} + \mathcal{L}_{Inter} \tag{29}$$

Thus, the final loss function used for training is:

$$\mathcal{L} = \mathcal{L}_{BPR} + \lambda_1 \mathcal{L}_{KG} + \lambda_2 \mathcal{L}_{regular}, \tag{30}$$

where λ_1 and λ_2 are hyperparameters that control the balance between the supervised, contrastive, and regularization components.

Algorithm 1 Training procedure of CRHCL.

Input: User-item interaction matrix Y, KG $\mathcal{G}$, number of epochs E, batch size B
Output: Trained prediction function $\mathcal{F}(u, i \mid \mathcal{G}, \Theta)$
Initialize model parameters Θ (e.g., embeddings, attention weights)
Extract ripple-based multi-hop triple sets for each u and i: $\mathcal{J}_{u_{init}}, \mathcal{J}_{u_{potential}}, \mathcal{J}_{i_{origin}}, \mathcal{J}_{i_{potential}}$ with Eq. (16)
Construct augmented triple views $\mathcal{J}_o^{l'}$ for CL
for epoch = 1 to E **do**
  Shuffle Y and split into mini-batches of size B
  for each mini-batch (u, i) sampled from Y **do**
    Generate negative samples j for each positive (u, i) pair
    Retrieve relevant triple sets for u, i, and sampled negatives
    Compute user and item embeddings from multi-layer propagation over $\mathcal{G}$
    if CL is enabled **then**
      Compute intra-layer contrastive loss $\mathcal{L}_{\text{Intra}}$ using augmented views with Eqs. (20) and (22)
      Compute inter-layer contrastive loss $\mathcal{L}_{\text{Inter}}$ across ripple levels with Eqs. (21) and (23)
    end if
    Predict score $\hat{y}_{u,i} = \mathcal{F}(u, i \mid \mathcal{G}, \Theta)$
    Compute total loss $\mathcal{L}$ with Eq. (30)
    Update parameters Θ via stochastic gradient descent with Eqs. (9)–(30)
  end for
  Evaluate model performance on validation and test sets
end for
return $\mathcal{F}(u, i \mid \mathcal{G}, \Theta)$

The training and optimization process of CRHCL is summarized in Algorithm 1. To enhance scalability and training efficiency, we apply negative sampling to construct mini-batches, which alleviates the computational burden and mitigates data sparsity issues. The model parameters are optimized using stochastic gradient descent by minimizing the overall loss function. Training continues until convergence or early stopping criteria are met. Detailed settings and sensitivity analysis of the hyperparameters will be provided in the experimental section.

4.6. Complexity and stability analysis

4.6.1. End-to-end time complexity

CRHCL's runtime is dominated by three modules. (1) Relation-aware semantic extraction over entity–relation interactions contributes $O(|\mathcal{E}| \cdot d)$, where $|\mathcal{E}|$ is the number of interactions (within a mini-batch) and d is the embedding dimension. (2) Dual-path propagation aggregates behavior-driven and KG-guided signals across L hops, adding $O(L \cdot d)$ per target (linear in depth). (3) Hierarchical contrastive learning (HCL) introduces intra-/inter-layer alignments with bounded positive/negative caps, yielding linear overhead in batch size and hop count. Under standard neighbor sampling, the overall complexity is well-approximated by $O(|\mathcal{E}| \cdot d + L \cdot d)$, preserving linear scaling in graph size and propagation depth.

4.6.2. Attention weight calculation

Let $S_{o,L}^l$ be the ripple (triple) set for target $o \in \{u, i\}$ at hop l, with cardinality $|S_{o,L}^l|$. The attention encoder $\pi(\cdot)$ is an MLP that consumes a fused d-dimensional input and projects to a hidden size h before scoring. The per-triple cost is $O(dh)$ for the first affine map plus $O(h)$ for activation/projection, while softmax costs $O(|S_{o,L}^l|)$. Thus the hop-wise complexity is $\text{Cost}_{\text{att}}(l) = O\big(|S_{o,L}^l| \cdot d \cdot h\big)$. For a deeper L_π-layer MLP, the exact bound is $O\big(|S_{o,L}^l| \cdot (dh + (L_\pi - 1)h^2)\big)$, with $L_\pi \in \{1, 2\}$ and $h \approx d$, $O(|S_{o,L}^l| \cdot d \cdot h)$ remains tight. Summing over hops yields $O\big(\sum_{l=1}^{L} |S_{o,L}^l| \cdot d \cdot h\big)$.

4.6.3. Contrastive objective complexity

For intra-layer contrast (Eqs. (20), (22)), if each anchor uses P_l positives and N_l negatives at hop l, the InfoNCE softmax costs $O\big((P_l + N_l) \cdot d\big)$ per anchor. Inter-layer contrast (Eqs. (21), (23)) has the same order as it pairs local layer-k views with global counterparts and treats other layers as negatives. Because P_l and N_l are capped by fixed neighbor limits, contrastive overhead scales linearly with batch size and hop count, and is negligible relative to backbone propagation.

4.6.4. Memory and scalability

The memory footprint comprises (i) embeddings for $|\mathfrak{I}|$ entities and $|\mathfrak{R}|$ relations: $O\big((|\mathfrak{I}| + |\mathfrak{R}|) \cdot d\big)$, (ii) mini-batch ripple buffers: $O(B \cdot L \cdot M \cdot d)$ with batch size B and per-hop neighbor cap K, and (iii) transient MLP activations: $O(B \cdot L \cdot M \cdot h)$. To avoid ripple explosion on large KGs, the implementation uses fixed-size neighbor sampling (M), shallow propagation ($L \leq 2$), cached attention scores for frequent triples, and mixed-precision training—keeping runtime and memory linear in the effective neighborhood size.

4.6.5. When complexity becomes limiting

In dense graphs where $|S_{o,L}^l|$ is large, the $O(|S_{o,L}^l| \cdot d \cdot h)$ term can dominate. Practical mitigations include: (1) top-k attention pruning before softmax, (2) importance sampling of triples using historical attention, (3) relation filtering to drop low-utility relations after a warm-up phase, and (4) precomputing per-relation projections to reduce $d \times h$ multiplications. These strategies bound the effective neighborhood around M and preserve near-linear scaling.

4.6.6. Stability in training

CRHCL integrates LayerNorm, LeakyReLU (to avoid dead units under sparse inputs), temperature $\tau = 0.2$ for well-conditioned contrastive gradients, gradient clipping, and early stopping. Empirically on MovieLens-1M, loss curves are smooth without gradient explosion/oscillation, and per-epoch time grows near-linearly with (L, M) and the number of interactions—consistent with the analysis above. These results indicate a balanced trade-off between semantic expressiveness and computational efficiency, together with stable optimization dynamics.

5. Experiment

5.1. Datasets

To comprehensively evaluate the effectiveness and generalizability of CRHCL, we conduct experiments on three widely used public benchmarks with diverse domains and characteristics: Last.FM,[1] Book-Crossing,[2] and MovieLens-1M.[3] These datasets differ significantly in scale, sparsity, and semantic richness, enabling a thorough examination of CRHCL under various data conditions.

We evaluate on three public benchmarks. Last.FM contains 1872 users, 3864 music items, and 42,346 interactions; its KG includes 9366

[1] https://grouplens.org/datasets/hetrec-2011/
[2] http://www.informatik.uni-freiburg.de/~cziegler/BX/
[3] https://grouplens.org/datasets/movielens/

Table 2
Statistical summary of the Last.FM and Book-Crossing datasets, along with the MovieLen-1M datasets.

Datasets	Last.FM	Book-Crossing	MovieLen-1M
# Users	1872	17,860	6,036
# Items	3864	14,910	2,445
# Interactions	42,346	139,746	753,772
# Entities (KG)	9366	77,903	182,011
# Relations (KG)	60	25	12
# Triples (KG)	15,518	151,500	1,241,996

entities, 15,518 triples, and 60 relation types, providing rich music-related semantics such as artists, albums, and tags. Book-Crossing consists of 17,860 users, 14,910 books, and 139,746 interactions with explicit ratings; its KG contains 77,903 entities, 151,500 triples, and 25 relation types, enabling fine-grained semantic modeling. MovieLens-1M comprises 6036 users, 2445 movies, and 753,772 interactions; its KG includes 182,011 entities, 12 relations, and 1,241,996 triples with metadata such as genres, directors, and actors, serving as a large-scale, dense benchmark to assess the scalability and robustness of CRHCL.

Overall, the three datasets cover distinct domains (music, books, movies) and different graph complexities, enabling a comprehensive and fair assessment of CRHCL across varying interaction patterns and semantic structures. Table 2 summarizes the key statistics of all datasets and their corresponding KGs.

5.2. Experimental settings

5.2.1. Evaluation metrics

To assess the recommendation performance of the proposed method, we evaluate it under two widely used paradigms: click-through rate (CTR) prediction and top-K recommendation. For both tasks, the user–item interaction data is randomly partitioned into training, validation, and test sets with a ratio of 6:2:2. All experiments are conducted independently on the Last.FM, Book-Crossing and MovieLens-1M datasets.

(1) CTR prediction. For the CTR prediction task, the model is trained to estimate the probability that a user interacts with an item. Evaluation is then performed on the test set by comparing predicted interaction probabilities with ground-truth labels. We employ two standard metrics: AUC (Area Under the ROC Curve), which reflects the probability that a randomly chosen positive instance ranks higher than a randomly chosen negative one. F1 score, which measures the balance between precision and recall, particularly suitable for imbalanced binary classification scenarios.
(2) Top-K recommendation. In this setting, the model is used to generate a ranked list of candidate items for each user based on predicted interaction scores. We report Recall@K, which measures the proportion of relevant items successfully retrieved within the top-K recommendations. Following common practice, K is set to 5, 10, 20, 50, and 100 to evaluate model performance at varying recommendation list lengths and granularities.

5.2.2. Baselines

To demonstrate the effectiveness of CRHCL, we compare it against a wide range of representative knowledge-aware recommendation models spanning five major methodological categories:

- **Collaborative filtering-based methods:** We include BPRMF (Rendle et al., 2012), a classic matrix factorization model optimized via Bayesian Personalized Ranking, serving as a strong CF-based baseline.
- **Embedding-based methods:** CKE (Zhang et al., 2016) is a representative model that combines collaborative filtering with KG embeddings (e.g., TransE).
- **Path-based methods:** PER (Yu et al., 2014) utilizes meta-path-based semantic relevance and attention to capture personalized preferences. RippleNet (Wang et al., 2018) adopts ripple network propagation to simulate the hierarchical influence of knowledge paths on user interests.
- **GNN-based methods:** KGCN (Wang et al., 2019b) applies neighborhood aggregation over the KG to generate entity-aware representations. KGAT (Wang et al., 2019c) further introduces attention mechanisms over entities and relations. COAT (Dai et al., 2022) improves contrastive signal propagation in knowledge-aware environments via attentive message passing.
- **CL-based methods:** We include KGCL (Yang et al., 2022) and CMCLRec (Xu et al., 2024), two recent CL-based KG recommendation models that adopt graph-level contrastive objectives to improve representation robustness and enhance semantic discrimination. In addition, we incorporate GraphCL (Hafidi et al., 2020) and SimGCL (Yu et al., 2022) as additional baselines. GraphCL (Hafidi et al., 2020) serves as a general graph CL framework that maximizes the agreement between augmented graph views through structure- and feature-level perturbations. SimGCL (Yu et al., 2022), on the other hand, introduces a simplified contrastive regularization strategy without explicit augmentation, providing a lightweight and efficient paradigm for self-supervised recommendation. Including these two methods enables a more comprehensive comparison across both knowledge-enhanced and general CL paradigms.

These methods are selected for their representativeness, strong performance in prior studies, and wide adoption in knowledge-aware recommendation benchmarks. They cover diverse modeling paradigms, enabling a comprehensive and fair comparison with our proposed framework.

5.2.3. Parameter settings

In our experiments, all models are implemented using the PyTorch framework (Zhao et al., 2023). The learning rate is set to 5×10^{-3} across both datasets. We use the AdamW optimizer (Xie & Li, 2024) for stable convergence and better generalization, with a fixed batch size of 2048. Model parameters are initialized using the Xavier initializer (Glorot & Bengio, 2010), which helps maintain the variance of activations across layers during training.

For regularization, we set the L_2 weight decay coefficient to 10^{-4}, and the KG embedding regularization term to 10^{-6}. The embedding dimension of entities and relations in the KG is fixed to 64.

To capture multi-hop semantic dependencies, we construct ripple sets with a maximum of L layers (default $L = 3$), where the number of sampled neighbors per layer is selected from $\{8, 16, 32, 64, 128, 256\}$ depending on model complexity and dataset scale.

For all baseline models, we use the best-performing hyperparameter settings as reported in their original papers or determined via grid search on the validation set. This ensures a fair and consistent comparison across all methods.

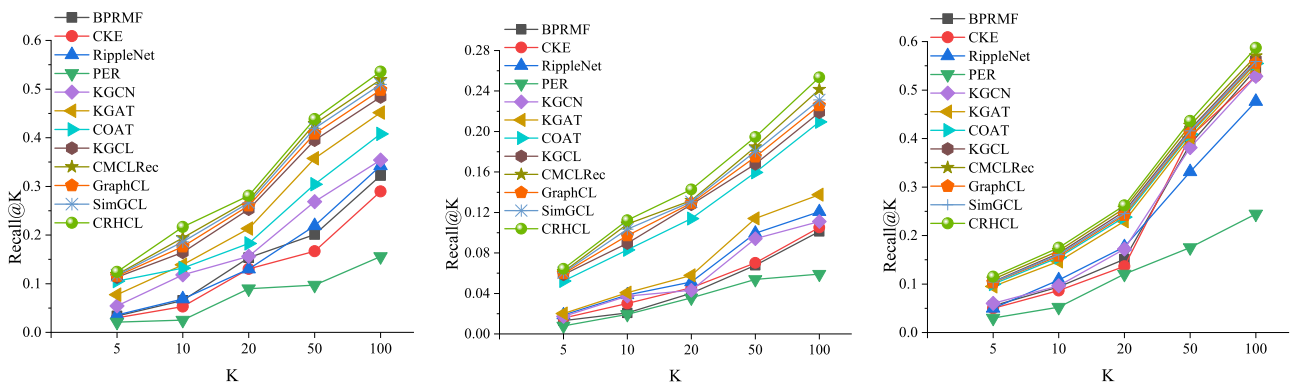
5.3. Performance comparison

Table 3 and Fig. 3 report the results on click-through rate (CTR) prediction and top-K recommendation tasks. Overall, the proposed **CRHCL** consistently achieves the best performance across all datasets (Book-Crossing, Last.FM, and MovieLens-1M).

CTR prediction. Across all three benchmarks (Table 3), CRHCL attains the best AUC/F1: 0.7863/0.7001 on Book-Crossing, 0.8675/0.7844 on Last.FM, and 0.9368/0.8633 on MovieLens-1M. Taking the strongest baseline (CMCLRec (Xu et al., 2024)) as reference, CRHCL delivers absolute gains of +0.74 %/+0.75 % on Book-Crossing, +0.72 %/+0.51 % on Last.FM, and +0.86 %/+0.54 % on MovieLens-1M. These consistent improvements on a larger and denser dataset further demonstrate the generalization and scalability of CRHCL.

Table 3
Results of AUC and F1 in CTR prediction.

Model	Book-Crossing		Last.FM		MovieLen-1M	
	AUC	F1	AUC	F1	AUC	F1
BPRMF (Rendle et al., 2012)	0.6583	0.6117	0.7563	0.7010	0.8920	0.7921
CKE (Zhang et al., 2016)	0.6759	0.6235	0.7471	0.6740	0.9065	0.8024
RippleNet (Wang et al., 2018)	0.7211	0.6472	0.7762	0.7025	0.9190	0.8422
PER (Yu et al., 2014)	0.6048	0.5726	0.6414	0.6033	0.7124	0.6670
KGCN (Wang et al., 2019b)	0.6841	0.6313	0.8027	0.7086	0.9090	0.8366
KGAT (Wang et al., 2019c)	0.7314	0.6544	0.8293	0.7424	0.9140	0.8440
COAT (Dai et al., 2022)	0.7440	0.6800	0.8217	0.7499	0.9163	0.8491
KGCL (Yang et al., 2022)	0.7453	0.6679	0.8455	0.7596	0.9184	0.8437
CMCLRec (Xu et al., 2024)	0.7789	0.6926	0.8603	0.7793	0.9282	0.8579
GraphCL (Hafidi et al., 2020)	0.7528	0.6732	0.8514	0.7643	0.9123	0.8524
SimGCL (Yu et al., 2022)	0.7635	0.6817	0.8562	0.7718	0.9176	0.8541
CRHCL	**0.7863**	**0.7001**	**0.8675**	**0.7844**	**0.9368**	**0.8633**



(a) Last.FM (b) Book-Crossing (c) MovieLens-1M

Fig. 3. Recall@K results in top-K recommendation on three datasets (K = 5, 10, 20, 50, 100).

Top-K recommendation. Fig. 3 reports Recall@K results for $K = \{5, 10, 20, 50, 100\}$. CRHCL consistently yields the highest Recall@K across all benchmarks, including Book-Crossing, Last.FM, and MovieLens-1M. Its advantage becomes more prominent at larger K values, indicating stronger ranking capability and long-tail retrieval performance.

Performance analysis. CRHCL's superior performance stems from two core designs. First, the context-aware relation perception module models entity–relation interactions along heterogeneous paths, effectively disentangling contextual semantics and capturing subtle user intents. Second, hierarchical contrastive learning enforces semantic consistency both locally and globally, enhancing noise-resilience and generalization.

In comparison, earlier KG-based models (e.g., CKE (Zhang et al., 2016), RippleNet (Wang et al., 2018)) incorporate external knowledge but do not perform layered semantic calibration. GNN-based models (e.g., KGAT (Wang et al., 2019c), KGCN (Wang et al., 2019b)) aggregate high-order signals but lack explicit modeling of context-dependent relational semantics. CMCLRec (Xu et al., 2024) improves robustness via multi-scale CL but does not explicitly resolve relational polysemy.

5.4. Ablation study

5.4.1. Effect of activation function in relation-context encoding

To justify the choice of LeakyReLU in the relation-context encoding module, we conduct an ablation study comparing three commonly used activation functions: ReLU, GELU, and LeakyReLU. The results across the Book-Crossing, Last.FM, and MovieLens-1M datasets are presented in Table 4.

Across all datasets, LeakyReLU consistently outperforms both ReLU and GELU in terms of AUC and F1 scores. These improvements can be attributed to the characteristics of KG propagation, where entity and relation embeddings are typically sparse and unevenly distributed. ReLU, with its zero cutoff, tends to excessively suppress weak but meaningful signals, while GELU, although smoother, may retain noisy activations in ambiguous relational contexts. LeakyReLU, in contrast, preserves low-magnitude negative values, which ensures stable gradient flow and facilitates the capture of subtle relational semantics across multi-hop reasoning paths.

The performance gains observed on the large-scale MovieLens-1M dataset further demonstrate that LeakyReLU is beneficial in dense, real-world recommendation scenarios. This highlights its robustness and scalability, validating its suitability for relational representation learning in knowledge-enhanced recommendation settings.

5.4.2. Ablation study for CRHCL

To evaluate the individual contributions of the two core components in CRHCL, namely the context-aware relation modeling (CR) module and the hierarchical contrastive learning strategy (HCL) module, we conduct an ablation study using multiple simplified variants of the model:

- **CRHCL$_{\text{w/o HCL}}$**: This variant disables the multi-level CL mechanism, removing both intra-layer and inter-layer contrastive objectives.
- **CRHCL$_{\text{w/o CR}}$**: This variant eliminates the relation-aware semantic modeling module and instead applies uniform aggregation across all entity-relation paths, disregarding contextual semantics.
- **CRHCL$_{\text{w/o Inter}}$**: This variant removes only the inter-layer CL component while retaining intra-layer CL.
- **CRHCL$_{\text{w/o Intra}}$**: This variant removes only the intra-layer CL component while retaining inter-layer CL.

The performance comparison is shown in Fig. 4, covering both the Last.FM and Book-Crossing datasets, and further extended to the

Table 4
Performance comparison of different activation functions in the relation-context encoding module.

Activation	Book-Crossing		Last.FM		MovieLens-1M	
	AUC	F1	AUC	F1	AUC	F1
ReLU	0.7835	0.6978	0.8652	0.7821	0.9349	0.8615
GELU	0.7851	0.6992	0.8664	0.7835	0.9358	0.8624
LeakyReLU	**0.7863**	**0.7001**	**0.8675**	**0.7844**	**0.9368**	**0.8633**

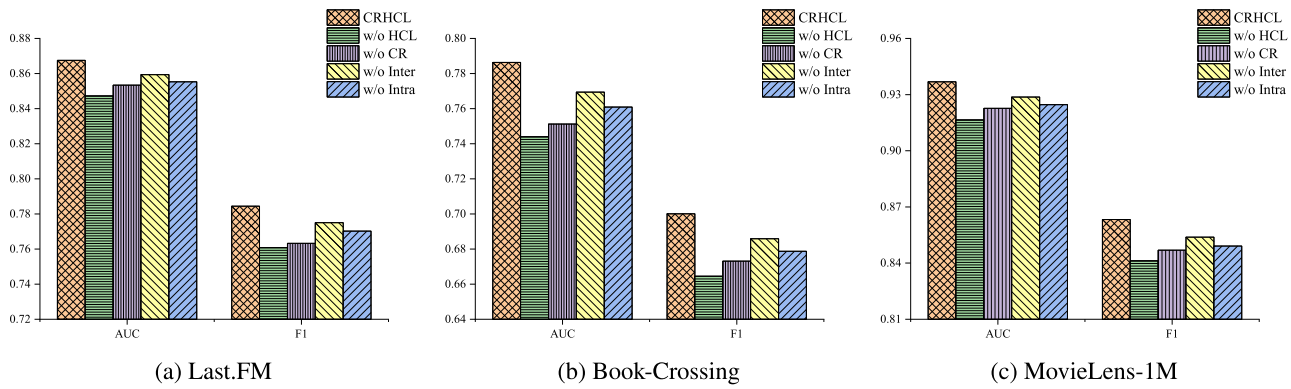


Fig. 4. Performance comparison of the full CRHCL model and its ablated variants (w/o HCL, w/o CR, w/o Inter, and w/o Intra) across three datasets.

MovieLens-1M dataset to verify scalability. The following observations can be drawn from the results:

- Excluding the CR module ($CRHCL_{w/o\ CR}$) leads to a significant performance decline in terms of both AUC and F1-score. This result emphasizes the importance of adaptive relation modeling, which enables the model to capture context-specific semantics by assigning different attention weights to relational paths. Without CR, the model struggles to distinguish relevant paths from noisy ones, leading to degraded representational quality.
- The removal of the HCL module ($CRHCL_{w/o\ HCL}$) also results in a consistent decrease in performance. HCL improves the model's robustness by aligning semantic representations across different layers and minimizing the impact of noisy information during propagation. Without this component, the learned embeddings become less stable and less discriminative.
- The fine-grained ablation reveals distinct roles of HCL subcomponents: removing intra-layer CL ($CRHCL_{w/o\ Intra}$) causes greater performance degradation than removing inter-layer CL ($CRHCL_{w/o\ Inter}$). This indicates that intra-layer consistency plays a more crucial role in capturing semantic relationships within the same propagation level, while inter-layer alignment provides complementary benefits for cross-level semantic coherence.

In conclusion, both components are essential to the effectiveness of CRHCL. The CR module enhances semantic extraction by incorporating relation-level awareness, while the HCL mechanism ensures representational consistency across hierarchical levels. The fine-grained analysis further demonstrates that intra-layer CL contributes more significantly to semantic consistency, while inter-layer CL provides additional robustness against noise propagation. Furthermore, the inclusion of MovieLens-1M results verifies that this observation holds in large-scale real-world environments, reinforcing the practical value and scalability of our hierarchical design. Together, these components significantly improve the model's accuracy, robustness, and generalization in knowledge-aware recommendation tasks.

5.5. Parameter sensitivity analysis

In this section, we conduct a sensitivity analysis to examine the impact of several key hyperparameters in the CRHCL model. The parameters considered include the aggregation function, model depth (i.e., the number of propagation layers), neighbor sampling size (M), and the temperature coefficient τ used in the CL module.

To ensure fairness and consistency, the tested values for each hyperparameter are selected based on commonly used configurations in mainstream knowledge-aware recommendation models (Wang et al., 2020b; Zou et al., 2022b). During the analysis, all other hyperparameters are held fixed at their optimal default values in order to isolate the effect of each individual factor on the model's overall performance.

5.5.1. Impact of aggregation function

To examine how aggregation functions affect performance, we compare three CRHCL variants that use Sum, Pooling, and Concat on Book-Crossing, Last.FM, and MovieLens-1M. Table 5 reports AUC and F1 scores on all three datasets. Across all benchmarks, Concat consistently achieves the highest scores. This superior performance can be attributed to Concat's ability to preserve layer-specific features, which is crucial for capturing the diverse characteristics of entity and relation embeddings at different levels. Unlike Sum and Pooling, which aggregate information in a way that may overlook important nuances, Concat ensures that multi-level signals are retained. This retention of detailed signals facilitates the construction of a more expressive final representation, capable of encapsulating richer semantics that are vital for effective recommendation systems (Luss et al., 2021; Wang et al., 2020a; Yen et al., 2017).

On MovieLens-1M, Concat again delivers the best AUC and F1, demonstrating that the advantage of this aggregation function holds even on large-scale, dense graphs. This indicates that the ability of Concat to retain multi-level interactions and relational semantics is particularly beneficial in complex datasets, where the relationships between entities are intricate and the amount of data is substantial. Thus, Concat's performance remains robust, supporting its scalability in real-world recommendation scenarios.

5.5.2. Impact of knowledge extraction depth

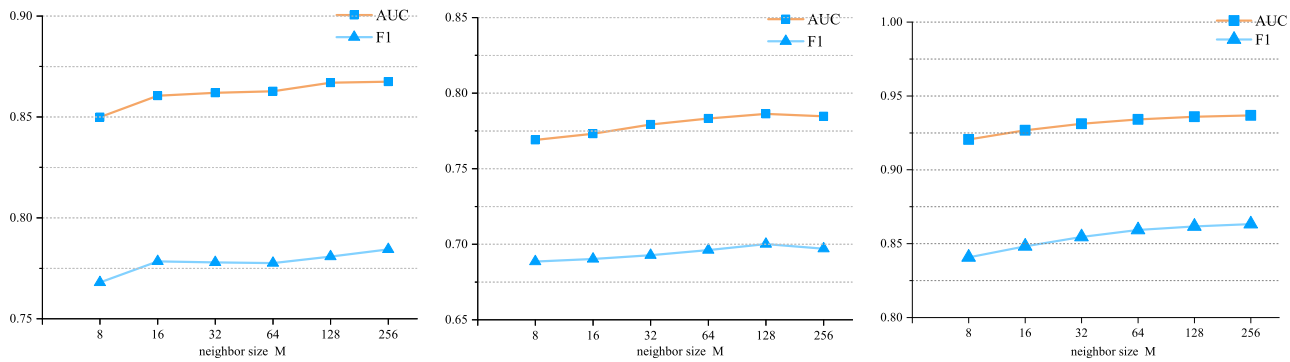
We analyze the effect of knowledge extraction depth by varying the number of propagation layers L from 1 to 3. The AUC and F1 results are reported in Table 6. On the Book-Crossing dataset, the model achieves the best performance at $L = 1$, while on Last.FM, optimal results are observed at $L = 2$. Similarly, on the larger MovieLens-1M dataset, $L =$

Table 5
Results of AUC and F1 for different aggregation functions.

Aggregator	Book-Crossing		Last.FM		MovieLens-1M	
	AUC	F1	AUC	F1	AUC	F1
Concat	**0.7863**	**0.7001**	**0.8675**	**0.7844**	**0.9368**	**0.8633**
Pool	0.7629	0.6824	0.8551	0.7783	0.9281	0.8552
Sum	0.7697	0.6890	0.8522	0.7755	0.9225	0.8498

Table 6
Results of AUC and F1 for different propagation depths L.

Propagation depth L	Book-Crossing		Last.FM		MovieLens-1M	
	AUC	F1	AUC	F1	AUC	F1
$L = 1$	**0.7863**	**0.7001**	0.8639	0.7813	0.9352	0.8621
$L = 2$	0.7838	0.6954	**0.8675**	**0.7844**	**0.9368**	**0.8633**
$L = 3$	0.7823	0.6959	0.8661	0.7832	0.9348	0.8615



(a) Last.FM (b) Book-Crossing (c) MovieLens-1M

Fig. 5. Effect of neighbor sampling size on model performance.

2 yields the highest performance, confirming that moderate multi-hop propagation effectively captures useful semantic signals while avoiding redundant noise.

This finding reflects the inherent trade-off in graph-based reasoning: deeper propagation expands semantic coverage and introduces richer relational cues, but excessive hops may accumulate irrelevant or noisy information, which weakens representation quality and harms prediction performance. For small and sparse datasets like Book-Crossing, shallow propagation is sufficient to capture meaningful semantics, while denser graphs such as Last.FM and MovieLens-1M benefit from a two-hop expansion that balances expressiveness and noise control.

Overall, these results underscore the importance of selecting an appropriate propagation depth. Setting $L = 2$ provides a robust default configuration for large-scale recommendation scenarios, achieving a balance between semantic enrichment and stability.

5.5.3. *Effect of neighbor sampling size*

We study how the neighbor sampling size in the ripple set impacts performance by varying the sampled neighbors per hop and reporting AUC/F1 in Fig. 5. On Last.FM, larger sampling generally improves results as more neighbors introduce richer high-order signals. On Book-Crossing, however, overly large sampling degrades performance due to noise from weakly relevant neighbors and increased training costs. This reflects the sensitivity to sparsity and the imbalance between initial and propagated entities. On MovieLens-1M, performance steadily improves as the sampling size increases, without the degradation observed on Book-Crossing. This suggests that denser user-item interactions and richer KG connectivity in MovieLens-1M provide stronger resistance to noise. The model benefits from the larger receptive fields enabled by higher sampling sizes, as the increased number of neighbors helps capture more complex relationships and interactions. In contrast to the sparse Book-Crossing dataset, MovieLens-1M's denser data structure facilitates more effective knowledge propagation, allowing for better utilization of larger neighbor sets.

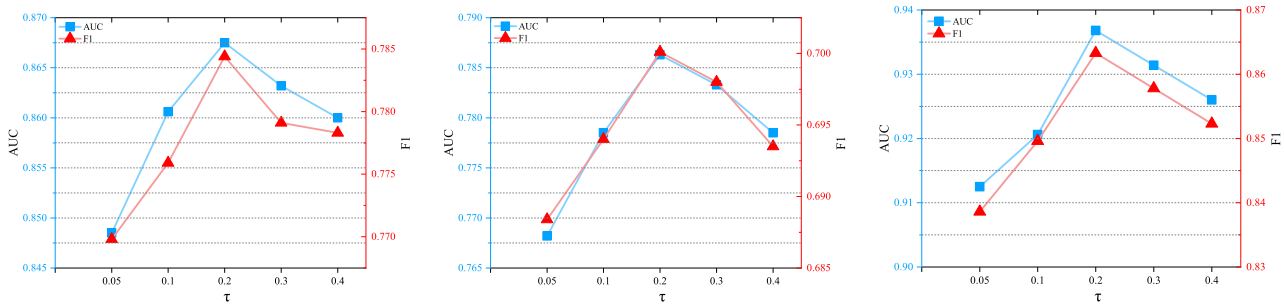
Overall, the number of sampled neighbors directly controls ripple density and the scale of triples, requiring careful tuning to balance information gain and computational overhead. For large-scale and dense datasets such as MovieLens-1M, larger sampling sizes are beneficial, as they enable the model to capture more comprehensive information. However, for sparse datasets like Book-Crossing, moderate sampling sizes are preferable to avoid introducing noise and excessive computational cost.

5.5.4. *Effect of the contrastive temperature coefficient*

The temperature coefficient τ in contrastive learning modulates the sharpness of the similarity-based softmax and thus the separation between positive and negative pairs (Wang et al., 2023b; Wu et al., 2021; Yang et al., 2024). Following Wang et al. (2021b), smaller τ amplifies penalties on dissimilar pairs, whereas larger τ softens contrasts. We sweep $\tau \in \{0.05, 0.1, 0.2, 0.3, 0.4\}$ on Book-Crossing, Last.FM, and MovieLens-1M and report AUC/F1 (Fig. 6).

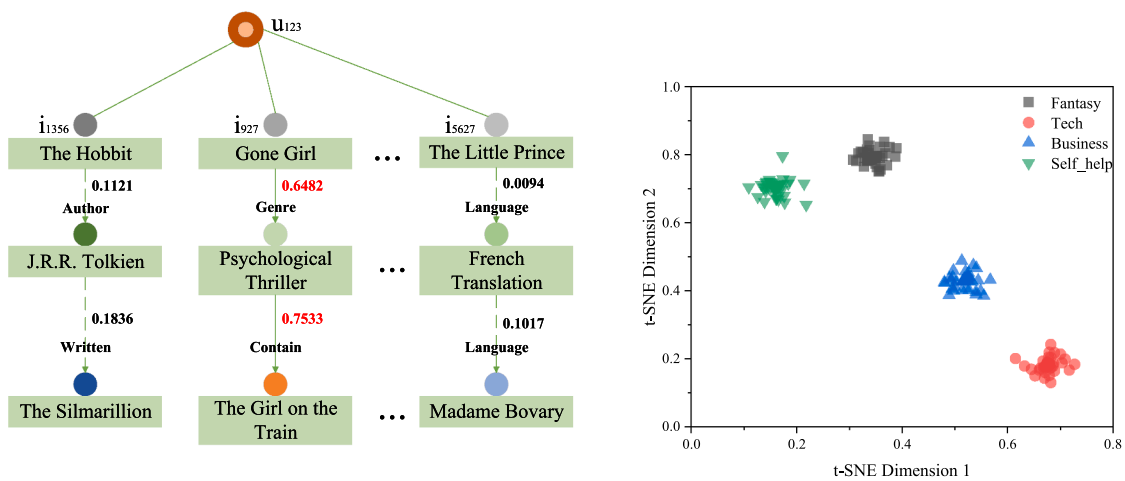
Across all three datasets, very small values of τ over-separate embeddings and negatively affect representation learning, as the model excessively penalizes dissimilar pairs. On the other hand, larger values of τ lead to under-discrimination, where positive and negative pairs are insufficiently differentiated. The optimal choice of τ is $\tau = 0.2$, which consistently strikes the best trade-off between penalty strength and semantic separability.

This trend also holds on the large-scale MovieLens-1M benchmark, where $\tau = 0.2$ provides a balance between maintaining meaningful separability and avoiding the over-penalization of dissimilar pairs. This indicates that a moderate temperature stabilizes contrastive gradients,



(a) Last.FM (b) Book-Crossing (c) MovieLens-1M

Fig. 6. Effect of temperature coefficient τ on contrastive learning performance.



(a) Attention-based relation visualization. (b) t-SNE projection.

Fig. 7. Interpretability analysis of CRHCL based on Book-Crossing.

leading to more effective learning by preventing both over-separation and under-discrimination, which is particularly important when dealing with the complex relationships present in large-scale datasets.

5.6. Case study

To further illustrate the interpretability of the proposed CRHCL framework, we conduct a qualitative case study on the Book-Crossing dataset. Specifically, we randomly select user u_{123} and visualize the model's attention distribution and semantic reasoning process in a knowledge-enhanced recommendation scenario. Fig. 7a displays a local knowledge subgraph connecting user u_{123} to multiple interaction items and candidate items through higher-order relationships (e.g., author, genre, language).

The visualization reveals that CRHCL effectively captures users' subtle semantic preferences by dynamically adjusting attention weights across different relationship paths. For instance, the attention score for the *Genre* relationship significantly exceeds weaker ties like *Author* or *Language*, indicating the model prioritizes semantic associations linked to user history and reading interests. Based on this reasoning pattern, CRHCL infers user u_{123} exhibits a strong preference for fantasy novels, aligning with the high-weight *Genre* path (*Psychological Thriller*). Consequently, the model recommends The Girl on the Train to the user, accompanied by the annotation: This book shares the same genre as titles you've read.

Fig. 7b presents a supplementary t-SNE visualization analyzing 200 randomly sampled entity embedding vectors from the Book-Crossing dataset, covering four major categories: *Fantasy*, *Tech*, *Business*, and *Self-Help*. The entity clusters learned by CRHCL exhibit clear semantic boundaries, with conceptually related entities (e.g., *The Hobbit*, *The Silmarillion*, and *The Lord of the Rings*) tightly clustered. This clustering pattern demonstrates that CRHCL not only enhances prediction accuracy but also generates semantically coherent embeddings aligned with human cognitive knowledge structures.

These results validate the model's interpretability from two complementary perspectives: (1) Attention visualizations intuitively reveal the most critical associative relationships influencing recommendations, (2) t-SNE plots confirm CRHCL captures semantically meaningful entity representations in the latent space. Together, they provide compelling qualitative evidence that CRHCL achieves an interpretable, context-aware knowledge-enhanced recommendation system.

6. Conclusion

This study presented CRHCL, a novel knowledge-aware recommendation framework designed to address two fundamental limitations in existing KG-based recommender systems: (1) static entity representations that fail to capture context-dependent semantic shifts across different relational paths, and (2) noise amplification caused by weak or irrelevant relations during multi-hop propagation. CRHCL integrates context-aware relation modeling to dynamically disambiguate entity semantics and a hierarchical contrastive learning strategy to enforce both intra-layer consistency and inter-layer alignment, thereby enhancing representation stability under noisy KG environments. By jointly modeling fine-grained contextual semantics and multi-level relational coherence, CRHCL effectively mitigates semantic ambiguity

and propagation-induced noise. Extensive experiments on three public benchmarks demonstrate that CRHCL consistently outperforms state-of-the-art methods in accuracy, semantic expressiveness, and robustness, while maintaining linear computational complexity and stable training behavior, confirming its scalability and practical viability.

Although CRHCL effectively captures contextual semantics and mitigates noise propagation in KG-enhanced recommendation, it still has two main limitations: it is built on static KGs without modeling temporal dynamics, and it relies on heuristic neighbor sampling, which may be suboptimal on large-scale graphs. In future work, we will extend CRHCL to streaming and temporally evolving KGs to dynamically update user intent and relation semantics, and design adaptive, budget-aware neighbor selection strategies that balance computational cost and representation quality for scalable real-time deployment.

CRediT authorship contribution statement

Quanchang Ye: Writing – original draft, Writing – review & editing, Visualization, Experimentation, Validation; **Xuxiang Feng:** Resources, Supervision, Writing – review & editing; **Hongzhen Xu:** Conceptualization, Formal analysis, Methodology, Writing – review & editing, Supervision; **Chao Wang:** Methodology, Formal analysis, Writing – review & editing; **Bao Zhang:** Data curation, Experimentation, Visualization; **Pengyang Wang:** Methodology, Validation, Writing – review & editing; **Chen Liu:** Formal analysis, Validation, Writing – review & editing; **Kafeng Wang:** Conceptualization, Formal analysis, Methodology, Writing – review & editing, Supervision.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

Bao, K., Zhang, J., Wang, W., Zhang, Y., Yang, Z., Luo, Y., Chen, C., Feng, F., & Tian, Q. (2025). A bi-step grounding paradigm for large language models in recommendation systems. *ACM Transactions on Recommender Systems*, *3*(4), 1–27.

Bordes, A., Usunier, N., Garcia-Duran, A., Weston, J., & Yakhnenko, O. (2013). Translating embeddings for modeling multi-relational data. *Advances in Neural Information Processing Systems*, *26*, 2787–2795.

Chen, Y., Yang, Y., Wang, Y., Bai, J., Song, X., & King, I. (2022). Attentive knowledge-aware graph convolutional networks with collaborative guidance for personalized recommendation. In *2022 IEEE 38th international conference on data engineering* (pp. 299–311).

Chen, Z., Gan, W., Wu, J., Hu, K., & Lin, H. (2025). Data scarcity in recommendation systems: A survey. *ACM Transactions on Recommender Systems*, *3*(3), 1–31.

Choi, M., Kim, H.-y., Cho, H., & Lee, J. (2024). Multi-intent-aware session-based recommendation. In *Proceedings of the 47th international ACM SIGIR conference on research and development in information retrieval* (pp. 2532–2536).

Corso, G., Stark, H., Jegelka, S., Jaakkola, T., & Barzilay, R. (2024). Graph neural networks. *Nature Reviews Methods Primers*, *4*(1), 17.

Dai, Q., Wu, X.-M., Fan, L., Li, Q., Liu, H., Zhang, X., Wang, D., Lin, G., & Yang, K. (2022). Personalized knowledge-aware recommendation with collaborative and attentive graph convolutional networks. *Pattern Recognition*, *128*, 108628.

De Min, T., Mancini, M., Alahari, K., Alameda-Pineda, X., & Ricci, E. (2023). On the effectiveness of layernorm tuning for continual learning in vision transformers. In *IEEE/CVF International conference on computer vision workshops* (pp. 3577–3586).

Deldjoo, Y., He, Z., McAuley, J., Korikov, A., Sanner, S., Ramisa, A., Vidal, R., Sathiamoorthy, M., Kasirzadeh, A., & Milano, S. (2024). A review of modern recommender systems using generative models (Gen-RECSYS). In *Proceedings of the 30th ACM SIGKDD conference on knowledge discovery and data mining* (pp. 6448–6458).

Elahi, E., Anwar, S., Al-kfairy, M., Rodrigues, J. J., Ngueilbaye, A., Halim, Z., & Waqas, M. (2025). Graph attention-based neural collaborative filtering for item-specific recommendation system using knowledge graph. *Expert Systems with Applications*, *266*, 126133.

Elfwing, S., Uchibe, E., & Doya, K. (2018). Sigmoid-weighted linear units for neural network function approximation in reinforcement learning. *Neural Networks*, *107*, 3–11.

Fan, W. (2024). Recommender systems in the era of large language models (LLMs). *IEEE Transactions on Knowledge and Data Engineering*, *36*, (6889–6907).

Glorot, X., & Bengio, Y. (2010). Understanding the difficulty of training deep feedforward neural networks. In *Proceedings of the thirteenth international conference on artificial intelligence and statistics* (pp. 249–256).

Hafidi, H., Ghogho, M., Ciblat, P., & Swami, A. (2020). GraphCL: Contrastive self-supervised learning of graph representations. *Computing Research Repository*, arXiv:2007.08025.

Hamilton, W., Ying, Z., & Leskovec, J. (2017). Inductive representation learning on large graphs. *Advances in Neural Information Processing Systems*, *30*, 1024–1034.

Hao, Y., Zhuang, F., Wang, D., Liu, G., Sheng, V. S., & Zhao, P. (2024). A general strategy graph collaborative filtering for recommendation unlearning. In *Proceedings of the 33rd ACM International conference on information and knowledge management* (pp. 799–808).

Hu, J., Hooi, B., He, B., & Wei, Y. (2025). Modality-independent graph neural networks with global transformers for multimodal recommendation. In *Proceedings of the AAAI conference on artificial intelligence* (pp. 11790–11798).

Huang, W., Han, A., Chen, Y., Cao, Y., Xu, Z., & Suzuki, T. (2024). On the comparison between multi-modal and single-modal contrastive learning. *Advances in Neural Information Processing Systems*, *37*, 81549–81605.

Jin, D., Yu, Z., He, D., Yang, C., Yu, P. S., & Han, J. (2021). Gcn for hin via implicit utilization of attention and meta-paths. *IEEE Transactions on Knowledge and Data Engineering*, *35*(4), 3925–3937.

Kim, M.-S., Jang, Y.-J., & Sung, T.-E. (2025). Graph-based technology recommendation system using GAT-NGCF. *Expert Systems with Applications*, *288*, (128240).

Lai, K.-H., Yang, Z.-R., Lai, P.-Y., Wang, C.-D., Guizani, M., & Chen, M. (2024). Knowledge-aware explainable reciprocal recommendation. In *Proceedings of the AAAI conference on artificial intelligence* (pp. 8636–8644).

Li, S., Guo, H., Tang, X., Tang, R., Hou, L., Li, R., & Zhang, R. (2024a). Embedding compression in recommender systems: A survey. *ACM Computing Surveys*, *56*(5), 1–21.

Li, Z., Wang, J., Chen, Z., Wu, K., Wei, Y., & Huang, H. (2024b). Adaptive graph neural networks for cold-start multimedia recommendation. In *2024 IEEE International conference on data mining* (pp. 201–210).

Liang, W., Meo, P. D., Tang, Y., & Zhu, J. (2024). A survey of multi-modal knowledge graphs: Technologies and trends. *ACM Computing Surveys*, *56*(11), 1–41.

Lin, Z., Tian, C., Hou, Y., & Zhao, W. X. (2022). Improving graph collaborative filtering with neighborhood-enriched contrastive learning. In *Proceedings of the ACM web conference* (pp. 2320–2329).

Liu, X., Lochman, Y., & Zach, C. (2023a). Gen: Pushing the limits of softmax-based out-of-distribution detection. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 23946–23955).

Liu, Z., Wan, G., Prakash, B. A., Lau, M. S. Y., & Jin, W. (2024). A review of graph neural networks in epidemic modeling. In *Proceedings of the 30th ACM SIGKDD conference on knowledge discovery and data mining* (pp. 6577–6587).

Liu, Z., Xu, Z., Jin, J., Shen, Z., & Darrell, T. (2023b). Dropout reduces underfitting. In *International conference on machine learning* (*vol. 202*), (pp. 22233–22248).

Luss, R., Chen, P.-Y., Dhurandhar, A., Sattigeri, P., Zhang, Y., Shanmugam, K., & Tu, C.-C. (2021). Leveraging latent features for local explanations. In *Proceedings of the 27th ACM SIGKDD conference on knowledge discovery & data mining* (pp. 1139–1149).

Meng, F., Meng, Z., Jin, R., Lin, R., & Wu, B. (2025). Doge: Llms-enhanced hyper-knowledge graph recommender for multimodal recommendation. In *Proceedings of the AAAI conference on artificial intelligence* (pp. 12399–12407).

Pan, S., Luo, L., Wang, Y., Chen, C., Wang, J., & Wu, X. (2024). Unifying large language models and knowledge graphs: A roadmap. *IEEE Transactions on Knowledge and Data Engineering*, *36*(7), 3580–3599.

Rendle, S., Freudenthaler, C., Gantner, Z., & Schmidt-Thieme, L. (2012). BPR: Bayesian personalized ranking from implicit feedback. *Proceedings of the Twenty-Fifth Conference on Uncertainty in Artificial Intelligence*, (pp. 452–461).

Sun, M., Xu, Y., Tan, Z., You, D., & Chen, Z. (2025). Multi-level graph contrastive learning for cold-start recommendation in mashup development. *Information Sciences*, *717*, (122319).

Sung, C., Kim, W., An, J., Lee, W., Lim, H., & Myung, H. (2024). Contextrast: Contextual contrastive learning for semantic segmentation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 3732–3742).

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, *30*, 5998–6008.

Wang, H., Zhang, F., Wang, J., Zhao, M., Li, W., Xie, X., & Guo, M. (2018). RippleNet: Propagating user preferences on the knowledge graph for recommender systems. In *Proceedings of the 27th ACM International conference on information and knowledge management* (pp. 417–426).

Wang, H., Zhang, F., Zhang, M., Leskovec, J., Zhao, M., Li, W., & Wang, Z. (2019a). Knowledge-aware graph neural networks with label smoothness regularization for recommender systems. In *Proceedings of the 25th ACM SIGKDD International conference on knowledge discovery & data mining* (pp. 968–977).

Wang, H., Zhao, M., Xie, X., Li, W., & Guo, M. (2019b). Knowledge graph convolutional networks for recommender systems. In *The world wide web conference* (pp. 3307–3313).

Wang, J., Shi, Y., Yu, H., Wang, X., Yan, Z., & Kong, F. (2023a). Mixed-curvature manifolds interaction learning for knowledge graph-aware recommendation. In *Proceedings of the 46th international ACM SIGIR conference on research and development in information retrieval* (pp. 372–382).

Wang, K., Gao, X., Zhao, Y., Li, X., Dou, D., & Xu, C.-Z. (2020a). Pay attention to features, transfer learn faster CNNs. In *International conference on learning representations*.

Wang, K., Wang, P., & Xu, C. (2023b). Toward efficient automated feature engineering. In *IEEE 39th International conference on data engineering* (pp. 1625–1637).

Wang, X., He, X., Cao, Y., Liu, M., & Chua, T.-S. (2019c). KGAT: Knowledge graph attention network for recommendation. In *Proceedings of the 25th ACM SIGKDD International conference on knowledge discovery & data mining* (pp. 950–958).

Wang, X., Huang, T., Wang, D., Yuan, Y., Liu, Z., He, X., & Chua, T.-S. (2021a). Learning intents behind interactions with knowledge graph for recommendation. In *Proceedings of the web conference* (pp. 878–887).

Wang, X., Liu, N., Han, H., & Shi, C. (2021b). Self-supervised heterogeneous graph neural network with co-contrastive learning. In *Proceedings of the 27th ACM SIGKDD conference on knowledge discovery & data mining* (pp. 1726–1736).

Wang, X., Wang, D., Xu, C., He, X., Cao, Y., & Chua, T.-S. (2019d). Explainable reasoning over knowledge graphs for recommendation. In *Proceedings of the AAAI conference on artificial intelligence* (pp. 5329–5336).

Wang, Z., Lin, G., Tan, H., Chen, Q., & Liu, X. (2020b). Ckan: Collaborative knowledge-aware attentive network for recommender systems. In *Proceedings of the 43rd International ACM SIGIR conference on research and development in information retrieval* (pp. 219–228).

Wang, Z., Zhang, J., Feng, J., & Chen, Z. (2014). Knowledge graph embedding by translating on hyperplanes. In *Proceedings of the AAAI conference on artificial intelligence* (pp. 1112–1119).

Wu, J., Wang, X., Feng, F., He, X., Chen, L., Lian, J., & Xie, X. (2021). Self-supervised graph learning for recommendation. In *Proceedings of the 44th international ACM SIGIR conference on research and development in information retrieval* (pp. 726–735).

Xie, S., & Li, Z. (2024). Implicit bias of AdamW:l∞-norm constrained optimization. In *International conference on machine learning* (pp. 54488–54510).

Xu, J., Chen, Z., Yang, S., Li, J., Wang, H., & Ngai, E. C. H. (2025). Mentor: Multi-level self-supervised learning for multimodal recommendation. In *Proceedings of the AAAI conference on artificial intelligence* (pp. 12908–12917).

Xu, X., Dong, H., Qi, L., Zhang, X., Xiang, H., Xia, X., Xu, Y., & Dou, W. (2024). CMCLRec: Cross-modal contrastive learning for user cold-start sequential recommendation. In *Proceedings of the 47th International ACM SIGIR conference on research and development in information retrieval* (p. 1589–1598).

Yang, Y., Huang, C., Xia, L., & Li, C. (2022). Knowledge graph contrastive learning for recommendation. In *Proceedings of the 45th International ACM SIGIR conference on research and development in information retrieval* (pp. 1434–1443).

Yang, Z., Ding, M., Huang, T., Cen, Y., Song, J., Xu, B., Dong, Y., & Tang, J. (2024). Does negative sampling matter? A review with insights into its theory and applications. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *46*(8), 5692–5711.

Yen, I. E.-H., Lee, W.-C., Chang, S.-E., Suggala, A. S., Lin, S.-D., & Ravikumar, P. (2017). Latent feature lasso. In *International conference on machine learning* (pp. 3949–3957).

Yin, N., Wang, M., Chen, Z., De Masi, G., Xiong, H., & Gu, B. (2024). Dynamic spiking graph neural networks. In *Proceedings of the AAAI conference on artificial intelligence* (pp. 16495–16503).

Ying, R., He, R., Chen, K., Eksombatchai, P., Hamilton, W. L., & Leskovec, J. (2018). Graph convolutional neural networks for web-scale recommender systems. In *Proceedings of the 24th ACM SIGKDD International conference on knowledge discovery & data mining* (pp. 974–983).

Yu, J., Yin, H., Xia, X., Chen, T., Cui, L., & Nguyen, Q. V. H. (2022). Are graph augmentations necessary?: Simple graph contrastive learning for recommendation. In *SIGIR '22: The 45th international ACM SIGIR conference on research and development in information retrieval, Madrid, Spain, July 11, - 15, 2022* (pp. 1294–1303).

Yu, X., Ren, X., Sun, Y., Gu, Q., Sturt, B., Khandelwal, U., Norick, B., & Han, J. (2014). Personalized entity recommendation: A heterogeneous information network approach. In *Proceedings of the 7th ACM International conference on web search and data mining* (pp. 283–292).

Zangerle, E., & Bauer, C. (2022). Evaluating recommender systems: Survey and framework. *ACM Computing Surveys*, *55*(8), 1–38.

Zhang, B., Xu, H., Shuang, R., & Wang, K. (2025a). Heterogeneous information-based self-supervised graph learning for recommendation. *The Journal of Supercomputing*, *81*(4), 507.

Zhang, F., Yuan, N. J., Lian, D., Xie, X., & Ma, W.-Y. (2016). Collaborative knowledge base embedding for recommender systems. In *Proceedings of the 22nd ACM SIGKDD International conference on knowledge discovery and data mining* (pp. 353–362).

Zhang, H., Shen, X., Yi, B., Liu, J., & Xie, Y. (2025b). A plug-in critiquing approach for knowledge graph recommendation systems via representative sampling. In *Proceedings of the ACM on web conference* (pp. 322–333).

Zhao, X., Le Hellard, T., Eyraud-Dubois, L., Gusak, J., & Beaumont, O. (2023). Rockmate: An efficient, fast, automatic and generic tool for re-materialization in pytorch. In *International conference on machine learning* (pp. 42018–42045).

Zheng, L., Jing, B., Li, Z., Tong, H., & He, J. (2024). Heterogeneous contrastive learning for foundation models and beyond. In *Proceedings of the 30th ACM SIGKDD conference on knowledge discovery and data mining* (pp. 6666–6676).

Zheng, X., Ni, Z., Zhong, X., & Luo, Y. (2022). Kernelized deep learning for matrix factorization recommendation system using explicit and implicit information. *IEEE Transactions on Neural Networks and Learning Systems*, *35*(1), 1205–1216.

Zou, D., Wei, W., Mao, X.-L., Wang, Z., Qiu, M., Zhu, F., & Cao, X. (2022a). Multi-level cross-view contrastive learning for knowledge-aware recommender system. In *Proceedings of the 45th International ACM SIGIR conference on research and development in information retrieval* (pp. 1358–1368).

Zou, D., Wei, W., Wang, Z., Mao, X.-L., Zhu, F., Fang, R., & Chen, D. (2022b). Improving knowledge-aware recommendation with multi-level interactive contrastive learning. In *Proceedings of the 31st ACM International conference on information & knowledge management* (pp. 2817–2826).